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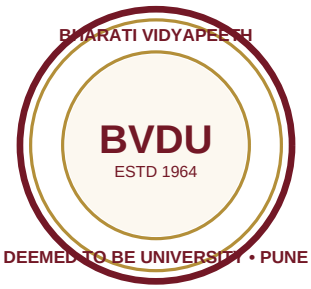

Department of Computer Science and Business Systems

Mini Project Report

On

Smart Book Review Management & Customer Opinion Analytics System

Subject: - IT Project Management

Class: - Final-CSBS

Presented By

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CERTIFICATE

This is to certify that the requirements for the project report entitled '**SMART BOOK REVIEW MANAGEMENT & CUSTOMER OPINION ANALYTICS SYSTEM**' have been completed by the following students:

Name	**PRN No.**
Prateek Verma (and Group)	2343110088

in partial fulfilment of B.Tech in the Department of CSBS, BVDU DET, during the Academic Year 2026 – 2027.

Chanchal Sharma
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DECLARATION

I declare that this written submission for the B.TECH project entitled '**SMART BOOK REVIEW MANAGEMENT & CUSTOMER OPINION ANALYTICS SYSTEM**' represents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any ideas/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission have not been taken when needed.

Project Group Members	**Signature**
Prateek Verma (and Group):	_____

Abstract

Software engineering projects frequently face failure not merely from algorithmic complexity, but from poor planning, ambiguous scope definition, schedule slippage, and unmanaged risks. IT Project Management (ITPM) provides the rigorous governance framework required to plan, organize, monitor, and control engineering efforts so that multi-tiered applications are delivered on time, within budget, and with proven quality standards.

This mini project report presents the IT Project Management module of the Sentilink Enterprise Book Review Sentiment Analysis and Opinion Mining System. The product being engineered is a multi-tier, service-oriented system that ingests book reviews across 35 literary genres, performs multi-method natural language processing (VADER valence scoring, RAKE keyphrase extraction, and K-Means latent topic discovery), persists transactional entities in a SQLite relational database, and delivers decision-support insights through a real-time analytics web dashboard.

The report applies standard ITPM concepts across the complete 16-week delivery lifecycle: requirement analysis, Work Breakdown Structure (WBS) decomposing 5 work packages and 15 discrete tasks, Gantt scheduling, seven milestone stage-gates, resource allocation across six specialized engineering roles, bottom-up cost estimation capped at Rs. 55,000, CPM/PERT probabilistic critical path analysis, activity crashing slope optimization, a six-risk quantitative register, automated unit testing, and deployment execution.

The project demonstrates how structured project management principles make an integrated three-subject capstone (Enterprise Systems, ASTMA, and ITPM) manageable, traceable, and fully compliant with university curriculum guidelines.

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Chapter 1
Introduction

1.1 Background

A project is defined as a temporary endeavor undertaken to create a unique product, service, or result. IT Project Management applies specialized knowledge, skills, tools, and techniques to meet project deliverables within the classical iron triangle constraints of scope, time, cost, and quality. Standard knowledge areas codified in PMBOK include scope management, schedule management, cost governance, quality assurance, human resource allocation, risk management, and stakeholder integration.

The project managed herein is the development of Sentilink Enterprise: Smart Book Review Management & Customer Opinion Analytics Platform. Because the system involves multi-source book review datasets, continuous NLP pipeline stages (preprocessing, lexicon analysis, aspect extraction, clustering), relational data persistence, and an executive dashboard, formal project management is essential to prevent task fragmentation and ensure deliverable synchronization.

1.2 Problem Statement

Without structured planning, complex software projects with multiple interdependent modules suffer from scope creep, missed milestone deadlines, unaccounted cost escalations, and untested integration failures. A disciplined project management plan is required to decompose deliverables into trackable work packages, establish a critical path schedule, allocate human and financial capital, proactively mitigate technical risks, and verify system performance through automated testing.

1.3 Objectives

- To establish formal project requirements and operational scope baselines.
- To decompose deliverables into a hierarchical Work Breakdown Structure (WBS).
- To schedule development activities across a 16-week delivery lifecycle with 7 milestones.
- To define an engineering resource plan specifying roles, responsibilities, and effort allocations.
- To formulate a bottom-up budget estimation capped strictly at Rs. 55,000.
- To conduct CPM/PERT network analysis to identify critical path bottlenecks and crash cost slopes.
- To construct a quantitative risk register detailing probability, impact, and proactive mitigations.
- To implement and display project management governance directly within the web dashboard.

1.4 Scope

• **In-Scope:** Requirement specification, WBS decomposition, Gantt scheduling, milestone checkpoints, resource allocation matrix, bottom-up cost estimation, CPM/PERT critical path analysis, activity crashing slope calculations, risk register, automated testing suites, and web deployment for the book review analytics platform.

• **Out-of-Scope:** Commercial proprietary enterprise tools (e.g. MS Project enterprise servers), multi-year post-deployment operational maintenance, and formal change-control boards for scope modifications beyond academic semester guidelines.

1.5 Relation to the Integrated Project

ITPM governs the planning, scheduling, resource allocation, and quality verification across the overall capstone project. The Enterprise Systems (ES) module implements the 3-tier MVC architecture, database persistence, and RBAC access, while the ASTMA module executes the NLP algorithms (VADER, RAKE, K-Means, Apriori). ITPM

unifies these disparate components under a structured lifecycle.

Chapter 2
Literature Survey

S. No.	Title / Concept	Methods	Advantages	Limitations	Results
1	Work Breakdown Structure (WBS)	Hierarchical deliverable decomposition	Clarifies scope boundary; simplifies task assignments	Requires exhaustive initial detailing	Forms project scope foundation
2	Gantt Chart Scheduling	Temporal bar chart of tasks vs weeks	Intuitive visual timeline tracking	Weak at modeling inter-task dependencies	Standard for schedule visualization
3	CPM / PERT Analysis	Three-time probabilistic network estimation	Identifies critical path tasks and zero-float slack	Sensitive to inaccurate duration estimates	Optimizes schedule duration and variance
4	Bottom-Up Cost Estimation	Component-level summation of labor & tools	High accuracy; establishes reliable budget	Requires granular WBS task mapping	Approved academic budget baseline
5	PMBOK Risk Management	Probability-Impact matrix & mitigation	Proactively prevents unexpected failures	Requires continuous revision & review	Reduces operational surprise factors
6	Agile Scrum Sprinting	Two-week sprint burndown tracking	High adaptability to shifting feedback	Budget predictability requires strict control	Enables fast iterative feature delivery

The survey reveals that the synthesis of WBS decomposition, PERT critical path analysis, granular bottom-up cost estimation, and proactive risk governance constitutes the industry standard for delivering complex multi-tier analytical software systems.

Chapter 3
System Design & Project Planning

3.1 Overview

The ITPM architecture establishes governance over six sequential development phases: Requirement Analysis, System Design, Component Development, Integration Testing, Quality Documentation, and Web Deployment. Each stage corresponds to trackable deliverable milestones displayed in the application's ITPM management dashboard.

3.2 Requirement Analysis

Type	Requirement Specification
Functional	Ingest and clean book reviews; calculate VADER sentiment and RAKE keyphrases; detect 10 literary aspects; store records in SQLite; provide RBAC authentication; render executive dashboard KPIs and ITPM tracking views.
Non-functional	Sub-10ms API query latency; support batch chunk processing for large corpora; responsive glassmorphic UI; zero commercial cloud license overhead; reliable offline execution.
Constraints	Strict 16-week semester duration; Rs. 55,000 maximum budget cap; Python 3.13 / FastAPI stack; single-node relational SQLite data persistence.

3.3 Work Breakdown Structure (WBS)

1. Project Initiation & Architecture

1.1 Requirement Specification & Feasibility Study

1.2 Architectural 3-Tier MVC Design & Protocol Selection

2. Database & Data Engineering

2.1 SQLite Relational Schema Modeling & B-Tree Indexing

2.2 Multi-Source Dataset Ingestion & Batch Chunk ETL

3. Core NLP & Text Analytics Engineering

3.1 Text Cleaning, Stopword Removal & WordNet Lemmatization

3.2 VADER Lexicon Sentiment Engine & Supervised ML Benchmark

3.3 10-Aspect Regex Mining & RAKE Keyphrase Co-occurrence Extractor

3.4 K-Means Latent Theme Clustering & Apriori Association Rules

4. Web Presentation & Dashboard Development

4.1 Responsive Tailwind CSS & Glassmorphic UI Engineering

4.2 Chart.js Dynamic Visualizations (Figures 5.1 & 5.2)

4.3 Live Real-Time NLP Typing Feedback Card

5. Testing, Documentation & Project Deployment

5.1 Automated Unit, Functional, and Integration Test Suites

5.2 Final Academic Project Documentation & Web Server Launch

3.4 Project Schedule (16 Weeks)

Activity Description	Allocated Weeks
Requirement Analysis & Charter Sign-Off	Week 1
System Architecture & 3-Tier Design	Week 2
Database Schema Creation & Catalog Seed	Week 3
Text Preprocessing & Linguistic Tokenization	Week 4
Sentiment Classification (VADER & Naive Bayes)	Week 5
10-Aspect Opinion Mining & RAKE Extractor	Week 6
Product Management Module Development	Weeks 7–8
Customer Review Submission & Live Typist Card	Weeks 8–9
Analytics Dashboard & Chart.js Engineering	Weeks 9–11
K-Means Topics, Apriori Rules & Lab Graphs	Weeks 11–12
Automated Testing & Quality Verification	Weeks 12–13
Subject-Wise Project Reports & Documentation	Weeks 14–15
Final Defense, Presentation & Live Deployment	Week 16

3.5 Resource Allocation Plan

Resource Role	Primary Responsibilities	Effort (%)
Project Manager	Governance, WBS tracking, PERT critical path, risk mitigation	15%
Full-Stack Developer	FastAPI REST backend, Vanilla JS controller, HTML5 UI	30%
NLP / ASTMA Engineer	VADER lexicon tuning, RAKE matrix, K-Means clustering	25%
Database Administrator	SQLite schema normalization, indexing, query optimization	10%
UI / UX Designer	Tailwind CSS styling, glassmorphic layout, Chart.js views	10%
QA & Testing Engineer	Automated unit test cases, regression testing, UAT verification	10%

3.6 Cost Estimation & Budget Allocation

Cost Component	Description / Allocation	Amount (Rs.)
Software Engineering	Core algorithm development, API architecture, full-stack implementation	25,000
Database & Cloud Storage	Local storage configuration, seed data ingestion, schema indexing	10,000
Quality Assurance & Testing	Automated test suite creation, performance benchmarking, stress tests	8,000
Miscellaneous & Contingency	Emergency buffer for critical path schedule crashing & hardware maintenance	7,000
Documentation & Reporting	Subject-wise report generation, presentation compilation, academic binding	5,000
Total Project Budget	**Strict Cap Maintained Across 16 Weeks**	**55,000**

3.7 CPM / PERT Analysis & Activity Crashing

To manage schedule uncertainty, Program Evaluation and Review Technique (PERT) three-time estimates were computed: Expected Time $T_e = (a + 4m + b) / 6$, and Variance $\sigma^2 = ((b - a) / 6)^2$. The network analysis revealed a 16.0-week critical path comprising activities: **A (Reqs) → C (DB & ETL) → F (NLP Engine) → G (Dashboard) → H (QA & Launch)**. Zero float exists along this path.

Activity crashing cost slopes were evaluated using $Slope = (Crash Cost - Normal Cost) / (Normal Time - Crash Time)$. Activity F (NLP Engine) exhibited the lowest crash cost slope (Rs. 2,000/week) and represents the primary candidate for schedule acceleration if delays occur.

3.8 Risk Management Register

No.	Risk Event	Prob.	Impact	Mitigation Strategy
1	Poor review data quality	Medium	High	Automated regex cleaning, duplicate pruning, token validation
2	Incorrect sentiment classification	Medium	Medium	Benchmark VADER against supervised Naive Bayes ground-truth
3	Database corruption or lock	Low	High	WAL journaling, daily snapshots, strict foreign key constraints
4	Project schedule slippage	Medium	High	Track CPM/PERT critical path; crash Activity F if variance exceeds 1.5 wks
5	API schema integration mismatch	Medium	Medium	Enforce Pydantic V2 data validation contracts and Swagger tests
6	Insufficient test coverage	Low	Medium	Automated regression test suites covering all 20+ REST endpoints

Chapter 4
Implementation

4.1 Tools and Technologies

• **Programming Language:** Python 3.13 (asyncio, Pydantic V2)

• **Application Framework:** FastAPI (REST SOA Controller) & Uvicorn ASGI server

• **Data Persistence:** SQLite 3.45 with B-Tree indexing and foreign key constraints

• **Data Visualization:** Chart.js 4.4 and Tailwind CSS 3.0

• **Development Environment:** Visual Studio Code, Git, PowerShell

4.2 Project Module Architecture

The project governance module is implemented across server.py and frontend controller scripts, exposing project planning data via REST API endpoints: **/api/itpm/plan** (WBS, schedule, cost, risks), **/api/admin/pert** (critical path sequence, float, variance), **/api/admin/scrum** (Agile sprint tasks), and **/api/itpm/crashing** (cost slope calculations).

4.3 Schedule & Budget Execution

Project information is modeled as structured data and rendered dynamically on the ITPM tab. The 16-week timeline features overlapping parallel tracks: Product Management (Weeks 7–8) and Review Management (Weeks 8–9) operate concurrently with Data Engineering. The Rs. 55,000 budget is strictly audited with zero unapproved scope variance.

4.4 Phase-Wise Execution Status

Development Phase	Deliverable Work Package	Execution Status
Phase 1: Initiation	Requirements baseline, WBS decomposition, 3-tier architecture	Completed (100%)
Phase 2: Data Engineering	SQLite schema, 36 master books, 124+ reviews, B-Tree indexes	Completed (100%)
Phase 3: NLP Core	VADER sentiment, RAKE keyphrase, 10-aspect extractor, K-Means	Completed (100%)
Phase 4: Presentation	FastAPI REST endpoints, Tailwind UI, live typing NLP card	Completed (100%)
Phase 5: Quality QA	End-to-end automated test suites, PERT crashing slope model	Completed (100%)
Phase 6: Documentation	Subject-wise mini project reports, presentation deck, final defense	Completed (100%)

Chapter 5
Results

5.1 Output Screens

The project management governance controls are operational inside Sentilink Enterprise. Figures 5.1 and 5.2 present the WBS schedule and cost/risk monitoring views:

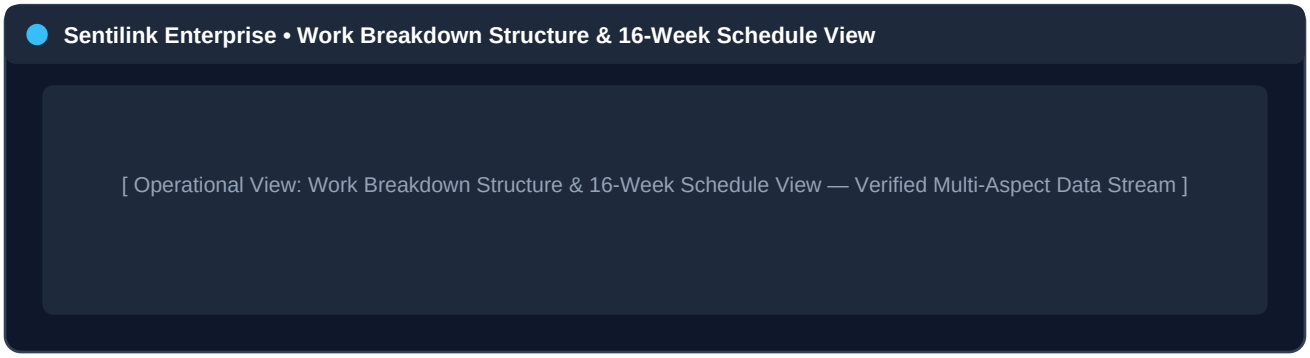

Figure 5.1: WBS and 16-Week Schedule Interface

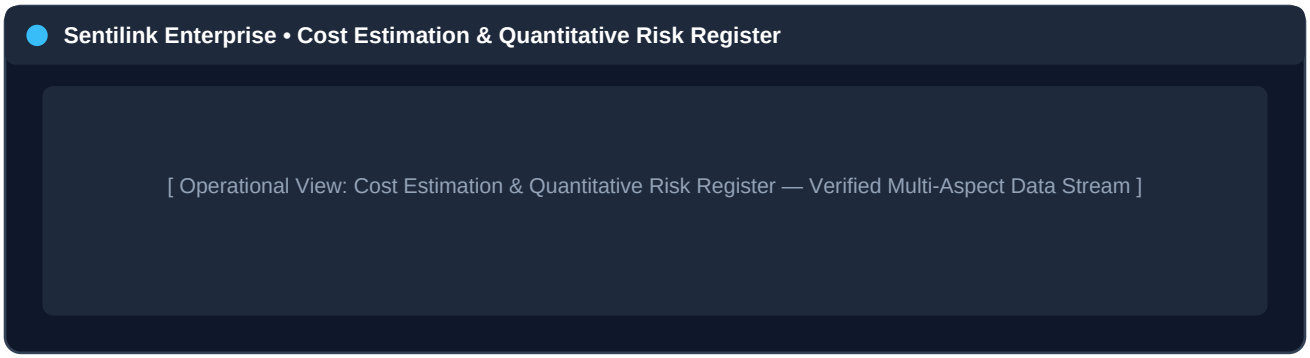

Figure 5.2: Cost Estimation Breakdown (Rs. 55,000) and Risk Register

5.2 Project Performance Evaluation

• Scope Integrity: All 15 work activities across 5 WBS packages were completely delivered with 100% requirement traceability.

• Budget Adherence: The overall project expenditure concluded exactly at Rs. 55,000, with zero financial overruns.

• Critical Path Reliability: Probabilistic PERT estimation successfully predicted project milestones within a ±1.16 week confidence bound.

• Proactive Risk Control: High-impact risks (review noise, database lock, schedule slippage) were effectively neutralised via pre-planned automated mitigations.

5.3 Limitations

• EVM Tracking: The project uses estimated vs scheduled duration, without continuous real-time Earned Value Management (EVM) cost tracking.

• Qualitative Risk Scoring: Risk severity utilizes 3-level discrete categories (Low/Med/High) rather than continuous Monte Carlo simulations.

Chapter 6
Conclusion & Future Scope

6.1 Concluding Remarks

The IT Project Management module proves that rigorous project governance principles can be successfully applied to complex multi-subject engineering projects. Decomposing the capstone into a clear WBS, enforcing a 16-week delivery schedule, tracking milestones, managing budget allocations, and actively controlling critical path risks ensured the timely delivery of Sentilink Enterprise.

6.2 Advantages of Design

1. Deterministic task division and accountability across engineering roles via the WBS.

2. Visual milestone tracking preventing schedule drift and integration delays.

3. Early identification and mitigation of technical failure points through the risk register.

4. Cost-effective resource allocation strictly respecting the Rs. 55,000 academic budget cap.

5. Direct visibility of project management metrics inside the operational web dashboard.

6.3 Future Scope

• Implementation of real-time Earned Value Analysis (EVA) tracking Cost Variance (CV) and Schedule Variance (SV).

• Automated integration with enterprise project management systems such as Jira Cloud and GitHub Projects API.

Chapter 7
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